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THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

ENGINEERING GEOLOGY AND ITS APPLICATIONS IN CIVIL ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			5 x2=10
			BL	CO	M
1.	a.	What are the basic causes of earthquake?	2	1	02
	b.	Name any two types of sedimentary rocks and provide an example of each.	2	2	02
	c.	List two mineras from the Quartz group and their industrial applications.	2	3	02
	d.	Why is understanding structural features crucial for dam construction?	2	4	02
	e.	Name any two types of aquifers and provide a characteristic of each.	2	5	02
		PART-B			6 X 5=30
2.	a.	What are the engineering problems commonly associated with earthquakes?	3	1	05
		OR			
	b.	Describe the composition of the Earth's internal structure based on seismic data.	3	1	05
3.	a.	How are rocks selected as construction materials?	3	2	05
		OR			
	b.	Explain the rock cycle with a neat sketch.	3	2	05
4.	a.	Discuss the special properties helping in identifying the ore minerals.	3	3	05
		OR			
	b.	Differentiate between Hematite and Magnetite.	3	3	05
5.	a.	Discuss the role of geotechnical investigations in assessing the stability of structures.	3	4	05
		OR			
	b.	Define a fault and explain its potential impact on construction.	3	4	05
6.	a.	Explain the method of geophysical investigations in groundwater exploration.	3	5	05
		OR			
	b.	Briefly explain the concept and methods of groundwater recharge.	3	5	05
7.	a.	Discuss the various components of the hydrological cycle and their	3	1	05

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		interrelationships.			
		OR			
	b.	Discuss the physical properties and uses of the Carbonate group of minerals.	3	3	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Explain the concept of plate tectonics and its significance in understanding earthquakes.	3	1	**10**
		OR			
	b.	What is the role of Engineering Geology in the field of Civil Engineering?	3	1	**10**
9.	**a.**	Explain the identification and description of metamorphic rocks such as quartzite, marble and slate. Discuss their applications in construction.	3	2	**10**
		OR			
	b.	How the sedimentary rocks are formed? Add a note on types of sedimentary rocks.	3	2	**10**
10.	**a.**	What are the physical properties commonly used to identify rock-forming minerals?	3	3	**10**
		OR			
	b.	Discuss the properties of quartz group minerals. Add a note on its varieties and uses in the industry.	3	3	**10**
11.	**a.**	Explain the classification of fault and effect of faults.	3	4	**10**
		OR			
	b.	Discuss the classification of folds. What are the engineering considerations of folds?	3	4	**10**
12.	**a.**	Discuss the techniques used in hydrological and geophysical investigations for effective groundwater exploration, including their advantages and limitations.	3	5	**10**
		OR			
	b.	Compare and contrast the different types of aquifers, highlighting their properties and significance for groundwater management.	3	5	**10**
13.	**a.**	Explain the process of metamorphism and the different types of metamorphism.	3	2	**10**
		OR			
	b.	Discuss the role of structural geology in site selection and design for dams, reservoirs, and highways, emphasizing the influence of lithological and structural features.	3	4	**10**